Progress in Additive Manufacturing
https://doi.org/10.1007/s40964-026-01912-4

RESEARCH

Finite element model for the description of intrinsic heat treatment induced aging in 18Ni300 maraging steel during direct laser powder deposition

Ivan Rodrigues[1] · Augusto Moita de Deus[1,2] · Célio Gabriel Figueiredo-Pina[1,3] · Amélia Almeida[1,4]

Received: 3 December 2025 / Accepted: 8 August 2026
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026

Abstract
Laser additive manufacturing techniques have received in recent times increased attention in the case of the production of customized, high-value components from high-strength, age-hardenable maraging steels. These alloys require cooling to form a carbon-free lath martensite (α′) phase that is then age-strengthened through precipitation at approximately 480 °C. Homogenization and aging are currently accomplished through costly post-manufacturing heat treatments that can, in principle, be replaced by the intrinsic heat treatment (IHT) effect of additive manufacturing, caused by repeated localized heating of the workpiece. This work aims to estimate the precipitation hardening time experienced by a thin wall of maraging steel during additive laser powder deposition at the rate of 1 single track overlap per minute and a unidirectional laser travel speed of 10 mm/s. For this, a heat transfer finite element deposition model is developed from in-situ workpiece temperature measurements and then the local thermal histories are used to interpret the microhardness profiles of such thin walls. The model can estimate the intrinsic aging time by considering a 380–580 °C aging temperature range found through differential thermal analysis. The hardness in the as-built condition and the estimated intrinsic aging time fit the established early-stage hardening kinetics law exponent for maraging steels, thereby validating the intrinsic age-hardening conditions that led to the IHT aging in thin walls.

Keywords Additive manufacturing · Maraging steel · Direct laser deposition · Intrinsic heat treatment · FEA modeling

1 Introduction

Maraging steels are used in high strength precision-made components produced through conventional methods [1] but new, cost-effective additive manufacturing (AM) processes are an emerging route to produce reliable maraging steel components. AM processes feature rapid localized heating and cooling that results in a range of microstructures and determines the final material properties. Currently, 18 wt%Ni-Co class maraging alloys are frequently used in additive manufacturing processes [2] such as laser powder deposition (LPD) and selective laser melting (SLM), in part because of the good weldability that is characteristic of a substitutional maraging alloy [3]. Regardless of cooling rate, these alloys acquire a carbon-free, soft, lath martensite (α′) microstructure at room temperature. Then, an aging heat-treatment at temperatures close to 480 °C can be used to strengthen the martensitic alloy, typically up to 2000 MPa in yield strength, with minimal distortion [4].

✉ Ivan Rodrigues
ivan.rodrigues@tecnico.ulisboa.pt

✉ Célio Gabriel Figueiredo-Pina
celio.pina@estsetubal.ips.pt

Augusto Moita de Deus
amd@tecnico.ulisboa.pt

Amélia Almeida
amelia.almeida@tecnico.ulisboa.pt

1 CeFEMA-Center of Physics and Engineering of Advanced Materials, Lisbon, Portugal

2 Department of Mechanical Engineering, Instituto Superior Técnico, Universidade de Lisboa, Lisbon, Portugal

3 Department of Mechanical Engineering, School of Technology of Setúbal, Polytechnic University of Setúbal, Setúbal, Portugal

4 Department of Chemical Engineering, Instituto Superior Técnico, Universidade de Lisboa, Lisbon, Portugal

Published online: 02 September 2026

Table 1 Composition of the substrate (W720) and powder (W722) maraging alloys (wt%)

	Ni	Co	Mo	Ti	Cr	Mn	Si	C	S	P	Fe
Substrate	18.5	9	5	0.7	–	0.1	0.05	0.005	–	–	Balance
Powder	17–19	8.5–10	4.5–5.2	0.8–1.2	0.25	0.15	0.1	0.03	0.01	0.01	Balance

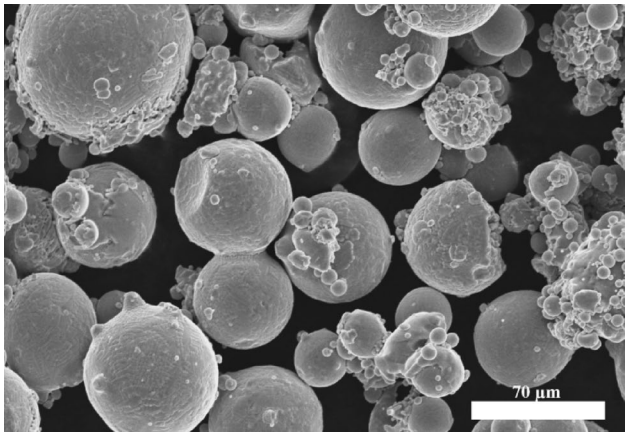


Fig. 1 Scanning electron microscope image of the feedstock powder morphology

This set of properties is put in use by industries such as tooling, aerospace and deep-sea operations, and applications include the fabrication of rocket-motor booster casings, helicopter rotor and jet turbine transmission shafts, aircraft landing and engagement components, pressure hulls, hydrofoils, dies and molds [1].

In LPD, a continuous track of added material is deposited by melting projected powder at the incidence spot of a high-power laser beam pointed at a substrate surface, and complex objects are then formed by overlapping consecutive deposition tracks. Hence, this process induces thermal effects on the already deposited tracks, due to the consecutive re-heating cycles caused by each deposition pass [5]. Previous studies have shown that the common LPD-fabricated 18 wt%Ni-Co maraging alloy can undergo age-hardening as a result of the induced intrinsic heat treatment (IHT), while the smaller melt pool and greater laser scanning speeds used in powder-bed SLM processes may hinder this effect [3, 6].

Thermal numerical modelling can estimate the temperature cycles and phase transformations that occur during SLM [7] and LPD [8] of maraging steels. These simulations can also predict the warping of an SLM component, the IHT aging caused by LPD for a set of aging criteria, and the heat accumulation phenomenon that occurs in the workpiece during SLM and LPD. Nonetheless, to the best of the authors' knowledge, there were no attempts to estimate IHT aging time based on a calibrated numerical model describing deposition temperatures.

This work is part of a study on the thermal effects of LPD and optimization of conditions leading to the IHT of maraging steels. This paper aims to estimate the precipitation time induced by the LPD fabrication process with 18Ni300 maraging steel, in comparison to resource-intensive post build-up heat treatments. For this, thin walls are built through unidirectional LPD of fully overlapping single-track layers deposited at 60 s intervals (Δt). An inverse finite element analysis (FEA) heat conduction model is developed from deposition temperature measurements at the base and atop of 30-layer (30L) and 60-layer (60L) walls. This model is used to calculate a transient temperature map of the wall during the deposition of the 30th and 60th layers, and this information is used, along with alloy phase transformation temperatures determined by differential thermal analysis (DTA), to produce microstructure maps for such walls. The maps are then evaluated relative to the hardness measured along the height of 30-layer and 60-layer walls for validation.

2 Materials and methods

The LPD experiments were carried out using powder and substrates of the 18Ni300 maraging steel grade, with similar nominal composition, as presented in Table 1. The substrates were milled plates of raw BÖHLER W720 alloy, with a thickness of 6 mm and surface sanded with P320 grit paper. The powder was BÖHLER W722 alloy (45–150 µm) with a relatively spherical morphology as observed in Fig. 1.

The LPD apparatus consisted of a high-power laser system, a powder feeder and a numerically controlled 2-axis horizontal table for substrate motion. The laser was a LASERLINE 1500 W LDM series diode laser, emitting radiation at 980 nm into a fiber optical system, leading to a beam spot of 3 mm at the surface of the substrate. Powder feeding was accomplished using a Plasma Technik Twin 10-C feeder, with argon as carrier and shielding gas, at a flow rate of 1.75 l/min. Powder feeding was done through a 2 mm diameter lateral nozzle.

Thin walls were deposited using a laser power of 1200 W, a powder flow rate of 0.145 g/s, and a laser scanning speed of 10 mm/s. The build-up strategy consisted of the sequential, unidirectional deposition of fully overlapping single tracks, stacked along the vertical build-up direction

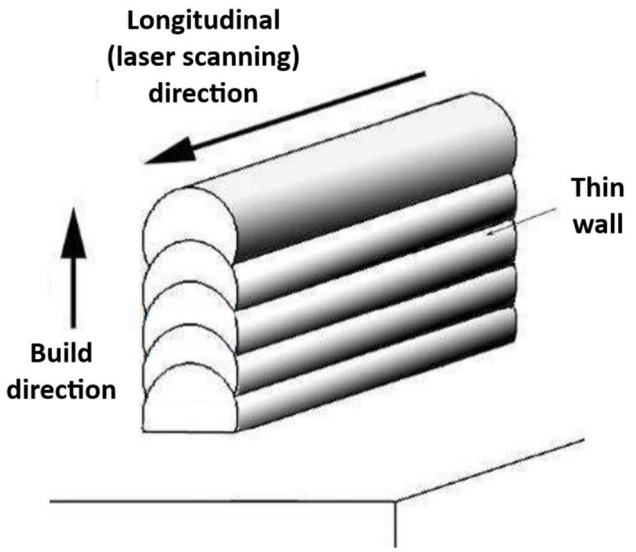


Fig. 2 Unidirectional deposition strategy for thin walls composed of overlayed layers, each with a height of 0.5 mm (adapted from [9])

as 0.5 mm thick layers (Fig. 2). Thin walls with 30 (30L) and 60 (60L) consecutive layers were deposited, with a time interval of 1 min between each layer ($\Delta t = 60$ s).

Some samples were submitted to post build-up heat-treatments consisting in solution annealing (SA) at 820 °C for 1 h and/or aging (A) at 480 °C for 3, 4.5 or 6 h, in a tubular furnace, under an argon gas atmosphere.

Table 2 lists all thin wall conditions and respective designations, regarding deposition strategy and post heat-treatments, done in sequence towards the rightmost columns.

A set of two infrared pyrometers were used to measure the temperatures at the midpoint of either the base or topmost track of a 60L wall during deposition. Each pyrometer measured a different temperature range at the same point of the wall: a high-temperature (HT) range (600–2300 °C) and a low-temperature (LT) range (−50–975 °C). Figure 3a illustrates the setting up of both devices used to measure the temperature at the midpoint of the base track during a first deposition instance. Figure 3b) shows how both pyrometers were instead raised vertically to match the height of the wall and capture the midpoint of each topmost track during a different deposition instance.

The HT pyrometer was a SENSORTHERM METIS M322, measuring a spot size of 1 mm in single-channel mode, using an emissivity of 0.9. The LT pyrometer was an OPTRIS CTfast LT15F, with the CF lens accessory,

Table 2 Samples designation according to build-up strategy and post deposition heat treatment sequence

Sample	Thin wall strategy	Solution annealing (SA)		Aging (A)	
	Height (#tracks)	Time [h]	Temperature [°C]	Time [h]	Temperature [°C]
Single track	1				
30 layers (30L)	30				
60 layers (60L)	60				
SA+A3h	30	1	820	3	480
A3h	30			3	480
A4.5 h	30			4.5	480
SA+A6h	30	1	820	6	480
A6h	30			6	480

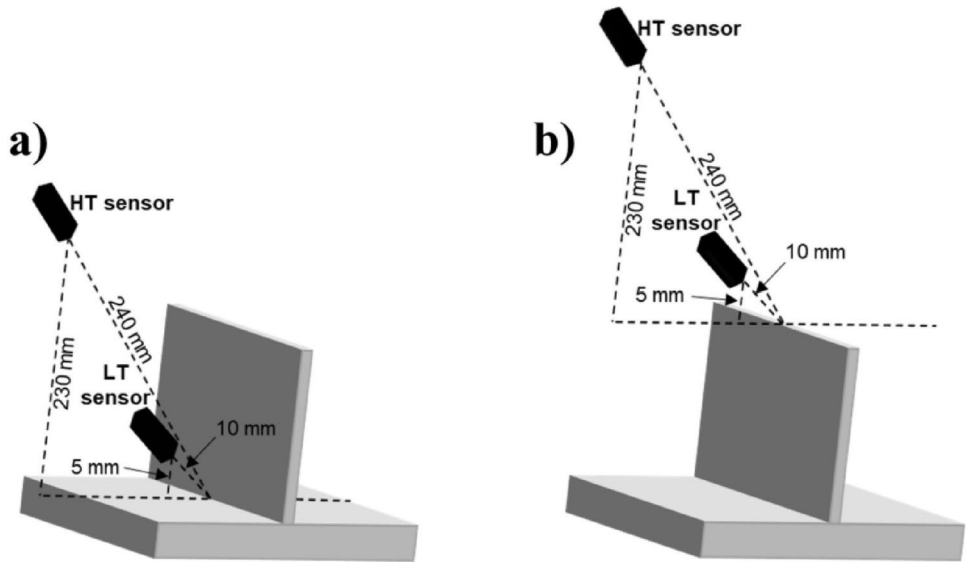


Fig. 3 Infrared pyrometer positions for in-situ temperature measurement during build-up of thin walls: **a** the set of pyrometers acquired the base layer temperature; **b** the set was raised to measure the temperature of each new layer

measuring a spot size of 2 mm using a single-color emissivity of 0.9. An alternative base temperature profile measurement method was implemented using a k-type thermocouple sensor fixed inside a 5-layer wall (5L) that served as substrate for the 60L wall. This sensor was embedded at the wall's longitudinal midpoint, 0.5 mm below the fifth layer.

The thin walls underwent metallographic preparation for subsequent microstructural characterization by optical microscopy and SEM/EDS and hardness measurements. Firstly, transverse cross-sections were cut from the deposition track at midpoint and mounted in resin. Then, the surface facing the end of the laser path was ground and polished to a 1 µm diamond suspension finish. SEM/EDS microanalysis required additional 0.1 µm diamond suspension polishing before etching in 5% Nital for 10 s. Micrographs were taken using an OLYMPUS CK40M optical microscope fitted with a digital camera. Microhardness measurements were done using a STRUERS Duramin-1 tester, with a Vickers indenter, under a load of 0.1 kg applied for 15 s. All hardness values are the average of four indentations measured in several regions, 0.25 mm apart.

DTA tests were carried out to determine the precipitation and phase transformation temperatures, using a Hitachi High-Tech STA7200 thermal analyzer, under a nitrogen atmosphere renewed at a flow rate of 200 ml/min. The tests were done using a heating cycle from ambient temperature to 940 °C, at 10 °C/min. The austenitic and martensitic transformation temperatures were determined during cooling at 30 °C/min from a 5-min isothermal stage at several temperatures.

FEA modeling using the SIMCENTER™ FEMAP™ software ver. 2021.2 MP1 and integrated NASTRAN solver were used to calculate the transient heat flow of two thin wall deposition cycles. The wall geometry mesh consisted of a thin rectangular slab with a length of 60 mm, a thickness of 2 mm and a height of either 15 or 30 mm to simulate the 30th or 60th layer, respectively. Both volumetric meshes were composed of 0.5 mm 10-node tetrahedral quadratic elements. The laser movement atop of each mesh was simulated using an estimated effective laser power, P_{eff}, applied as a nodal heat flux through a sequence of 30 adjacent 2 mm square surfaces. The initial temperatures of the base and topmost mesh planes were set according to the prior measurements, and the nodes in-between set to describe a linear temperature gradient. Then, the transient heat flow simulation began by fixing the base temperature at the measured 60 s cycle average and activating the laser heat flux at the first square. The entire 60 s deposition cycle simulations ran at a fixed calculation time step of 0.025 s.

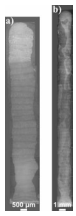


Fig. 4 Optical microscopy images of the etched transverse cross-sections of **a** 30-layer and **b** 60-layer single track walls built with Δt = 60 s

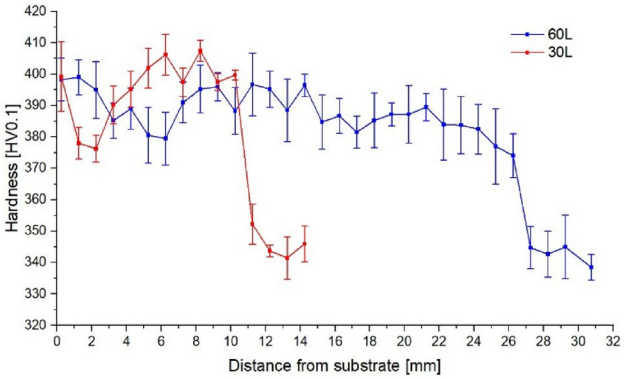


Fig. 5 Hardness variation along single-track walls as a function of the distance from the substrate

3 Results and discussion

3.1 Properties of multi-layer deposited structures

The optical micrographs in Fig. 4 show transverse cross-sections of multi-layer walls with 30 and 60 layers, taken at the respective longitudinal midpoint. The images show that both walls are continuous and free of macroscopic defects (pores or cracks) but slightly undulated with a narrow necking area near the base.

Figure 5 presents the hardness variation along the build direction of both walls, as a function of the distance from the substrate (vertical). It can be observed that both walls attain an average hardness value of 390 HV0.1, experiencing hardening.

The topmost regions of the walls present a hardness of about 340 HV0.1, that corresponds to the hardness measured for a single track (343 HV0.1). This suggests that in a region at the top of the thin walls with about 4 mm thickness, the material did not undergo sufficient reheating to promote hardening.

3.2 Intrinsic heat treatment assessment

The assessment of the conditions that lead to IHT hardening begins with the direct measurement by pyrometry of the 60L wall temperatures attained during deposition. This method employs an HT sensor to detect the temperature at the high thermal gradients zone near the melt pool, while the LT sensor captures the slower air-cooling step whereby

temperature tends to homogenize along the wall [5]. The data obtained from the HT and LT pyrometers is merged to obtain each of the plots presented in Fig. 6 which show the temperature cycles measured during deposition of a 60L wall, at the midpoint of the topmost and base layers as the red and blue lines, respectively.

Figure 6 shows that the topmost layer temperature increases throughout the deposition process, but the base remains at relatively lower temperatures, suggesting that the temperature at the base is quasi-stable from layer #20 onwards. The temperature reading from the thermocouple remains significantly lower than that of the top layer but slightly higher than that of the base layer, particularly after 40 min. The latter occurs because the sensor is placed within the topmost layer of a 5L wall, at a distance from the substrate, where some heat accumulation takes place.

For $\Delta t = 60$ s, the temperature at the top increases, keeping the material above the base temperature from the 4th layer onwards. The material is above 125 °C during cooling after the deposition of the 30th layer and above 165 °C before the deposition of the 60th layer.

The results of DTA tests allow to map the alloy transformation temperatures. The thermal analysis data presented in Fig. 7 reveals that four reactions can occur during the heating of the multi-layer wall: two distinct exothermic reactions with respective maximum rates at P_1 = 380 °C and P_2 = 580 °C; and two distinct endothermic reactions with respective maximum rates at N_1 = 676 °C and N_2 = 753 °C. Unlike the endothermic peaks, the width and intensity of each exothermic peak depend on the sample's thermal history. Cooling at 30 °C/min causes a single low-temperature exothermic reaction if the starting temperature is equal to or above the austenitic transformation start temperature A_s = 710 °C. If so, the cooling peak appears between a maximum temperature of M_s = 250 °C and a minimum temperature of M_f = 180 °C, the observed temperature range for the transformation.

Explanations for the exothermic reactions vary from martensite recovery to the precipitation and growth of coherent and incoherent particles [10, 11]. So, in this work several DTA heating and cooling curves were obtained for different heat-treatment conditions. The heating curves in Fig. 7 show that continued aging prior to DTA lowers the exothermic reaction rates at P_1 and P_2, which are:

- Maximum for the non-aged SA condition;
- Intermediate at the mid-height, IHT hardened zone of 30L walls (Fig. 5);
- Minimum at P_2 and undetectable at P_1 for the SA+A advanced aging condition.

Hence, the predominantly low IHT temperatures measured throughout the deposition (Fig. 6) indicate that:

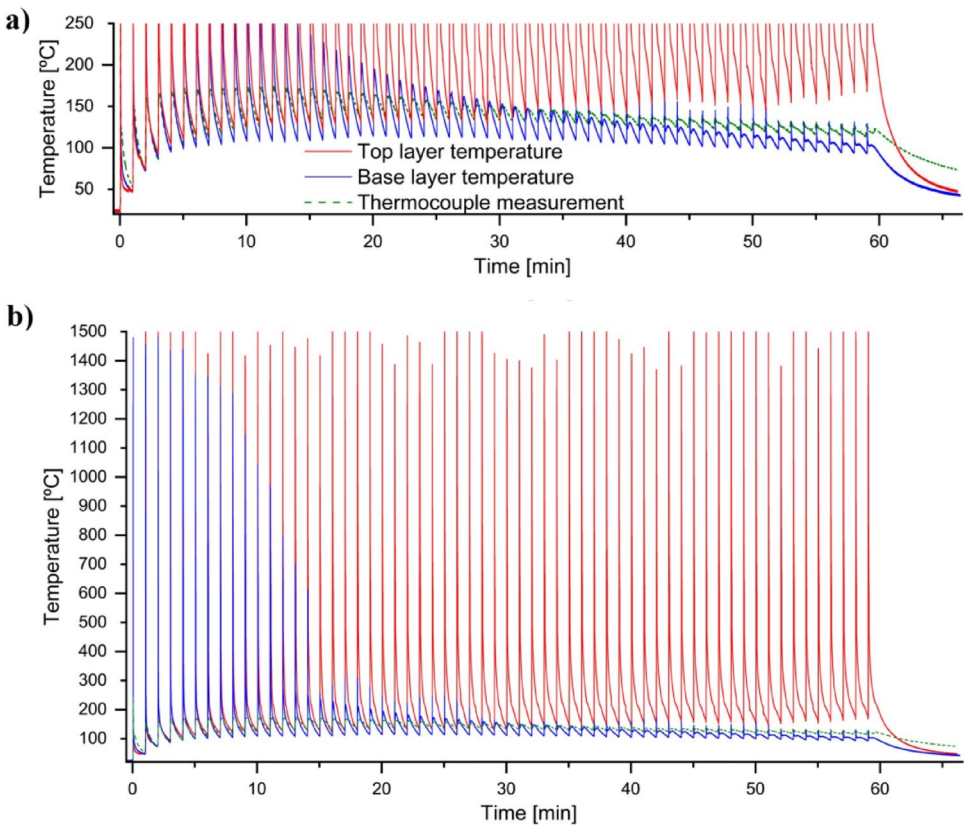


Fig. 6 Temperature profiles measured at the midpoint of the topmost (red line) and base (blue line) layers of a 60L wall during the build-up process: **a** shows temperatures up to 250 °C and **b** the full temperature range. From layer #4 onwards the base temperatures decrease relative to those at the topmost layer. The green dashed line shows the temperature of a thermocouple placed beneath the surface of a 5-layer deposit that served as substrate for a 60L wall

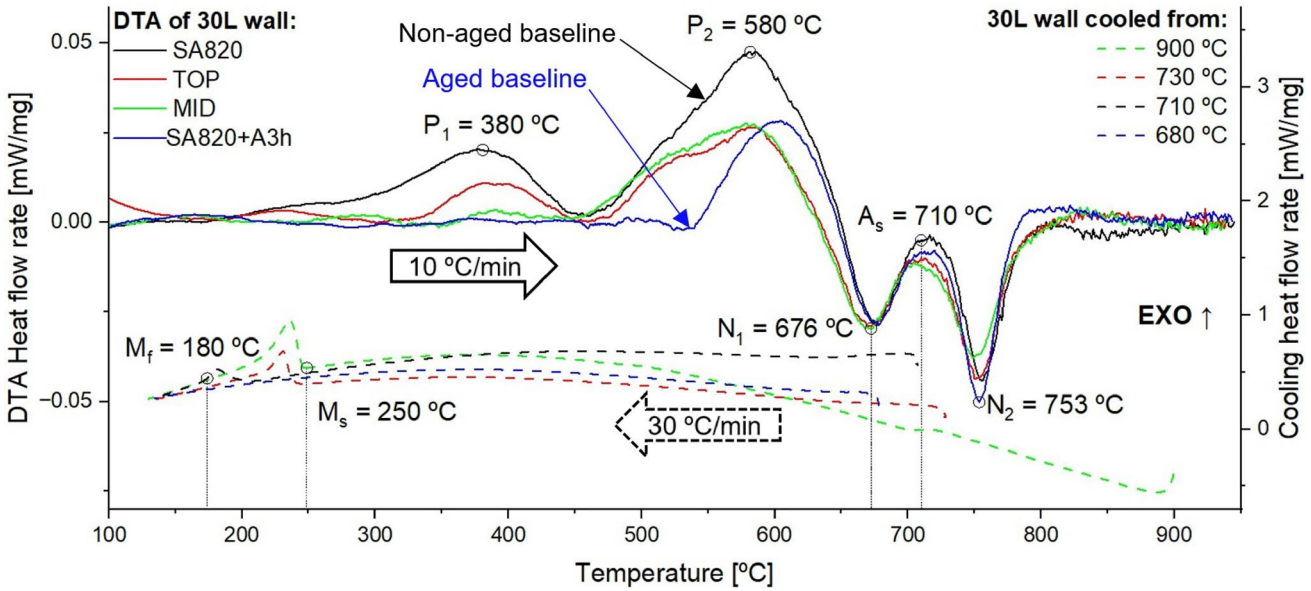


Fig. 7 Thermal analysis of wall segments built with Δt = 60 s. Solid lines represent DTA heating curves of as-deposited mid and top wall sections at 10 °C/min, and SA and SA+A the post heat treated wall; the dashed lines denote DTA cooling curves at 30 °C/min from isothermal stage at 680, 710, 730 or 900 °C

- The reaction at P_1 likely corresponds to the initial low temperature formation of coherent S+ω-phase [12, 13], which is completed through IHT at the hardened wall segment where subsequent DTA shows that reaction peak is suppressed;
- The reaction at P_2 likely corresponds to the formation of the main Ni_3(Ti,Mo) strengthening precipitates at higher temperatures [8, 12, 13], which is promoted mainly during SA+A because this treatment causes the greatest peak suppression during subsequent DTA.

The endothermic reactions at N_1 and N_2 are usually associated with austenite formation, but the cooling curves in Fig. 7 only show a martensitic transformation at M_s if the martensite is heated to at least 710 °C (the onset of the peak at N_2), hence:

- The reaction at N_2 likely corresponds to the complete transformation to austenite, which occurs at temperatures above A_s = 710 °C.
- The reaction at N_1 likely corresponds to a localized austenite formation due to Ni segregation (or "austenite reversion") following the redissolution of the main Ni_3(Ti,Mo) precipitates [8, 13] in favor of the formation of Fe-Mo phases [12, 14, 15].

3.3 Heat transfer model

A conduction finite element (FE) model of the built part is used to approximately describe the heat transfer conditions. Firstly, the simulation model is validated by fitting of the experimental measurements so that these could be accurately mapped during the deposition of two layers. A previous study on SLM reports experimental values for room-temperature density and temperature-dependent specific heat and thermal conductivity of an 18Ni300 alloy [16]. The temperature-dependent density function is taken from a separate study [7].

An effective laser power P_{eff} = 268 W accounts for the energy losses of a 3 mm circular laser spot illuminating a 2 mm thick wall with a reflectivity given by the Hagen-Rubens relation [17]. In addition, it is assumed for the alloy a 0.182 emissivity [18] and a value of 0.11 for the absorptivity [19], while the surrounding environment remains at 25 °C. Finally, two different convection heat transfer conditions are considered: the forced convection under the powder jet uses the h_{jet} coefficient and the remainder free-convection state uses h_{free}. The former is applied at temperatures above 600 °C using an average h_{jet} = 63.63 W m^{-2} K^{-1} taken from an analytical solution for Ar at 25 °C [20], and the latter h_{free} = 22 W m^{-2} K^{-1} shows the best fit to the experimental temperature profiles considering the typical values in literature [8]. This simulation can reasonably replicate the wall temperatures for layers #30 and #60, and the results are shown in Fig. 8, along with the simulated profiles of nodes situated 0.5 mm below the top of the mesh.

The simulations show that the 30L wall cools below M_f approximately 40 s after the laser interaction stopped, but the topmost layer of the 60L wall is kept at a comparatively higher temperature and requires 60 s to cool below

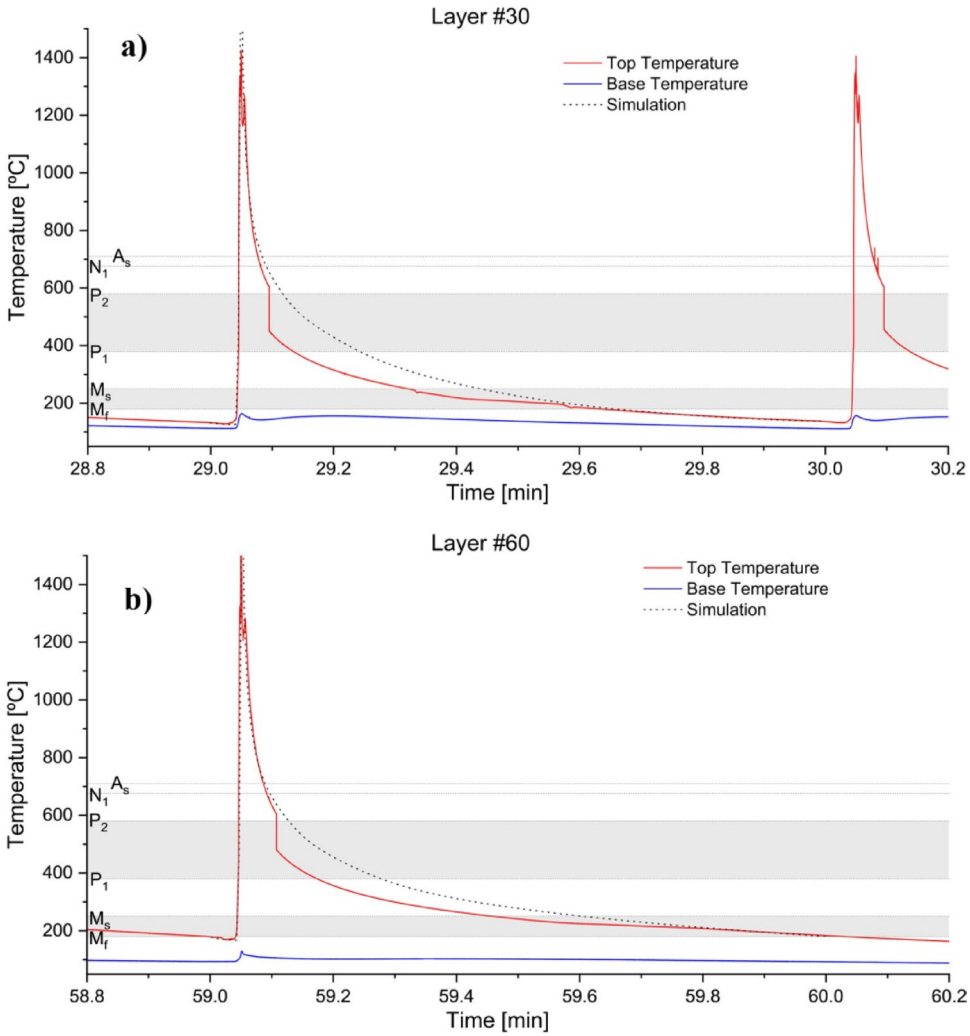


Fig. 8 Measured and simulated temperature profiles for the deposition of **a** the 30th and **b** the 60th layers of the walls with $\Delta t = 60$ s. Following laser interaction, cooling below the M_f temperature takes approximately 40 s in **a** and 60 s in **b**. The simulated temperature profiles of the top layer appear as dashed lines that fit the high and low gradient cooling phases

M_f despite its lower base temperature. The simulation also shows a reasonable fit to the top layer temperature profile during laser irradiation and subsequent cooling.

The FE generated temperature profile matches well the steady-state Rosenthal analytical solution [21] for the remainder of the domain between the topmost and base layers. Figure 9 compares the numerical solution at 5 s to the Rosenthal solution using workpiece temperature maps whereby each color-graded region is delimited by the closest reaction temperatures. The analytical model uses the same P_{eff}, a thermal conductivity of 23 W/(m K), a density of 8×10^3 kg/m^3, a specific heat of 460 J/(kg K) [22], a wall thickness of 2 mm, a laser speed of 10 mm/s and a wall temperature of either 405 K or 453 K to match the topmost layer temperatures measured at the start of the 30L and 60L wall deposition.

While the finite element model shows a reasonable fit to the experimental measurements and analytical solution, it does disregard some of the phenomena that occur experimentally. Figure 8 indicates that the measured temperature peaks widen significantly from layer #30 to #60, while the corresponding simulated profiles in Fig. 10 do not. This is likely due to some of the simplified assumptions used in the current model. One should expect the melt pool size to grow proportionally to wall height, because the cooling rates tend to decrease and the workpiece temperature rises [23], but this requires a proper description of the complex interaction of the laser beam with the traveling powder. Also, the enhanced convective heat transfer associated with the carrier gas is also simplified in the FEA, namely by considering a single convection coefficient along the wall, disregarding the turbulent flow along the wall surface. Finally, but most importantly, fluid flow in the melt pool is not considered. These simplifications of the model could be addressed in a fully coupled heat transfer and fluid flow analysis [24, 25] that was not attempted in the present study. The FE model developed in the current approach and its correlation to the analytical solution is intended as a straightforward, lab-ready, modelling framework to provide a validated tool to estimate aging times.

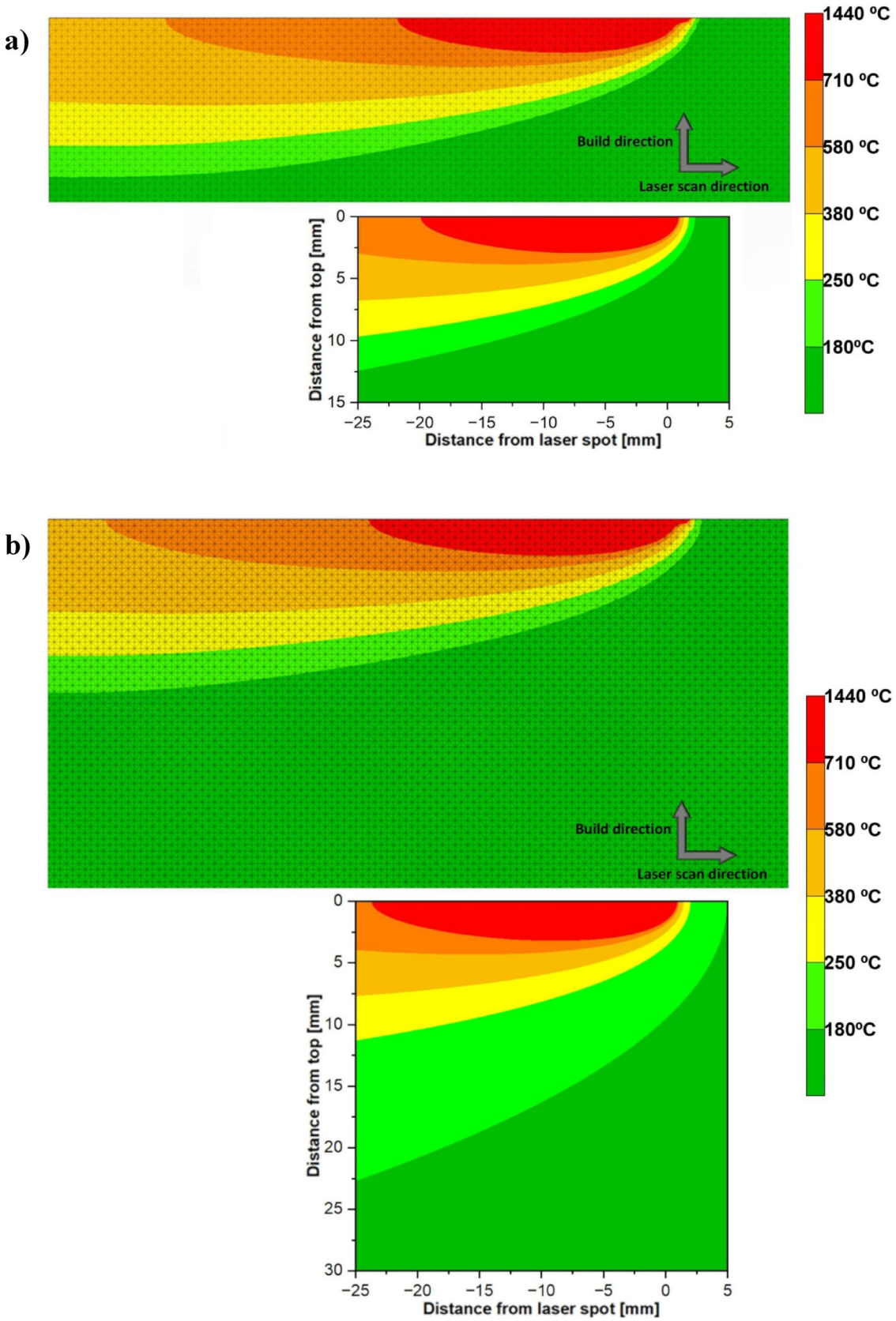


Fig. 9 Comparison of the wall temperature maps obtained for: **a** 30L and **b** 60L walls, showing the FE solution at 5 s above the corresponding steady-state Rosenthal solution

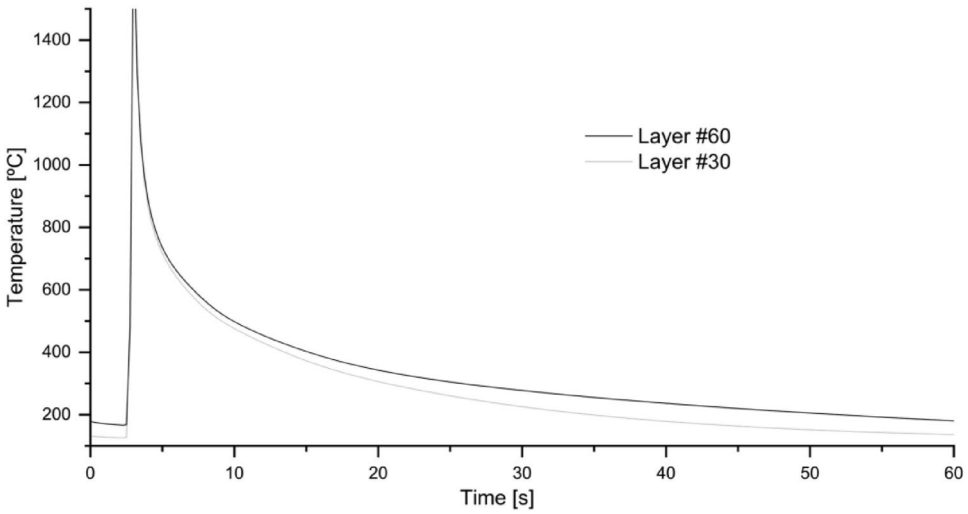


Fig. 10 Temperature profiles during deposition for layers #30 and #60, taken from a FEA node located 0.5 mm below the laser interaction surface. The laser interaction peak is slightly broader for layer #60

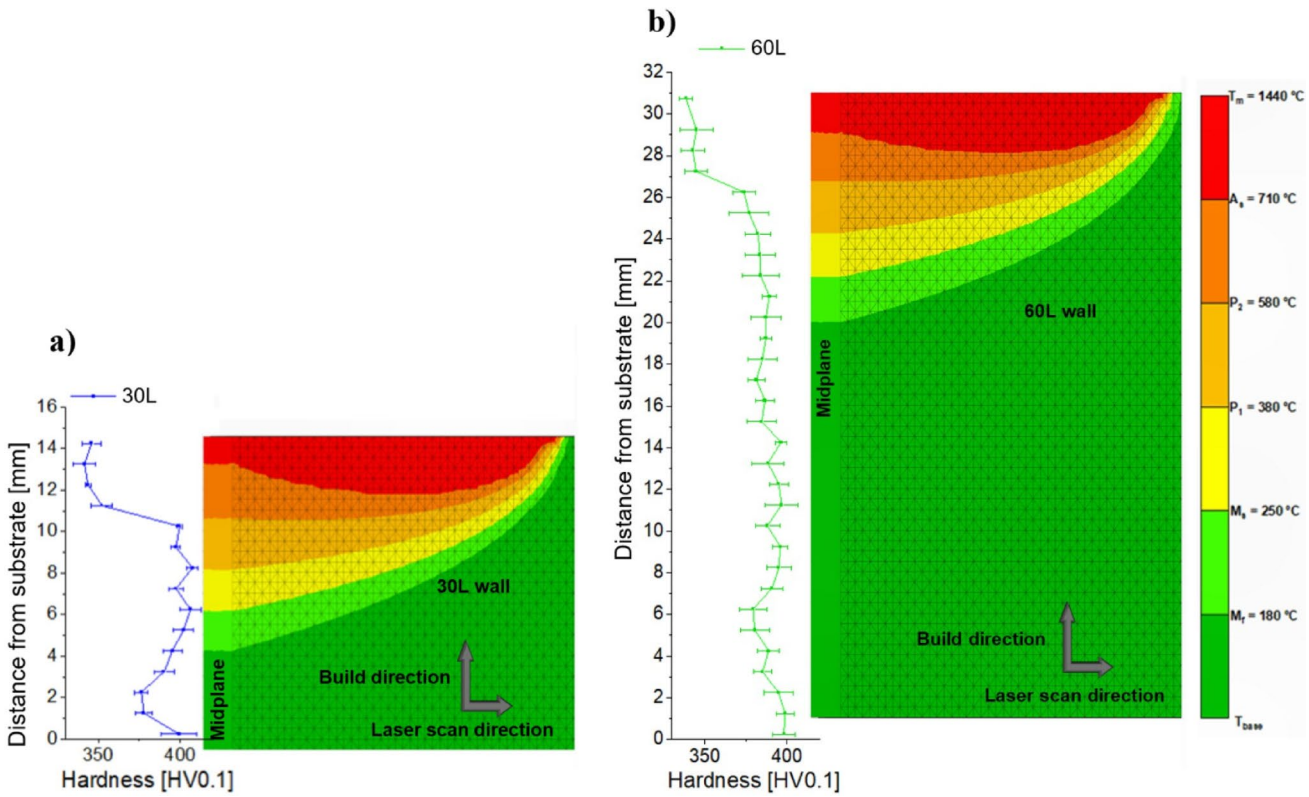


Fig. 11 FEA temperature map and as-built cross-section hardness of **a** 30L and **b** 60L walls built with a time between layers of 60 s. The sections are taken at the longitudinal midplane of the wall, and the FE solution is represented at 5 s after the laser activation. The hardened regions coincide with the wall domains that are kept below P_2 during the last deposition cycle

Despite the simplified approach, the FEA data, fed by the temperature readings, does correlate well with the sequence of transformations activated by the IHT induced during wall deposition and resulting hardness profile. Figure 8 shows that each new layer is allowed to cool below M_f enabling the formation of an age-hardenable α' phase [8], but the temperature maps in Fig. 11 suggest that a final reheating cycle above P_2 impairs IHT hardening. This latter effect agrees with the previous discussion in 3.2, as temperatures above P_2 are linked to the redissolution of hardening precipitates (at N_1 = 676 °C) and complete reaustenitization (at A_s = 710 °C).

The simulation results provide an estimate for the aging time experienced by the layers underneath the deposition surface. In this work, it is considered that aging occurs when an underlying layer is at a maximum temperature between

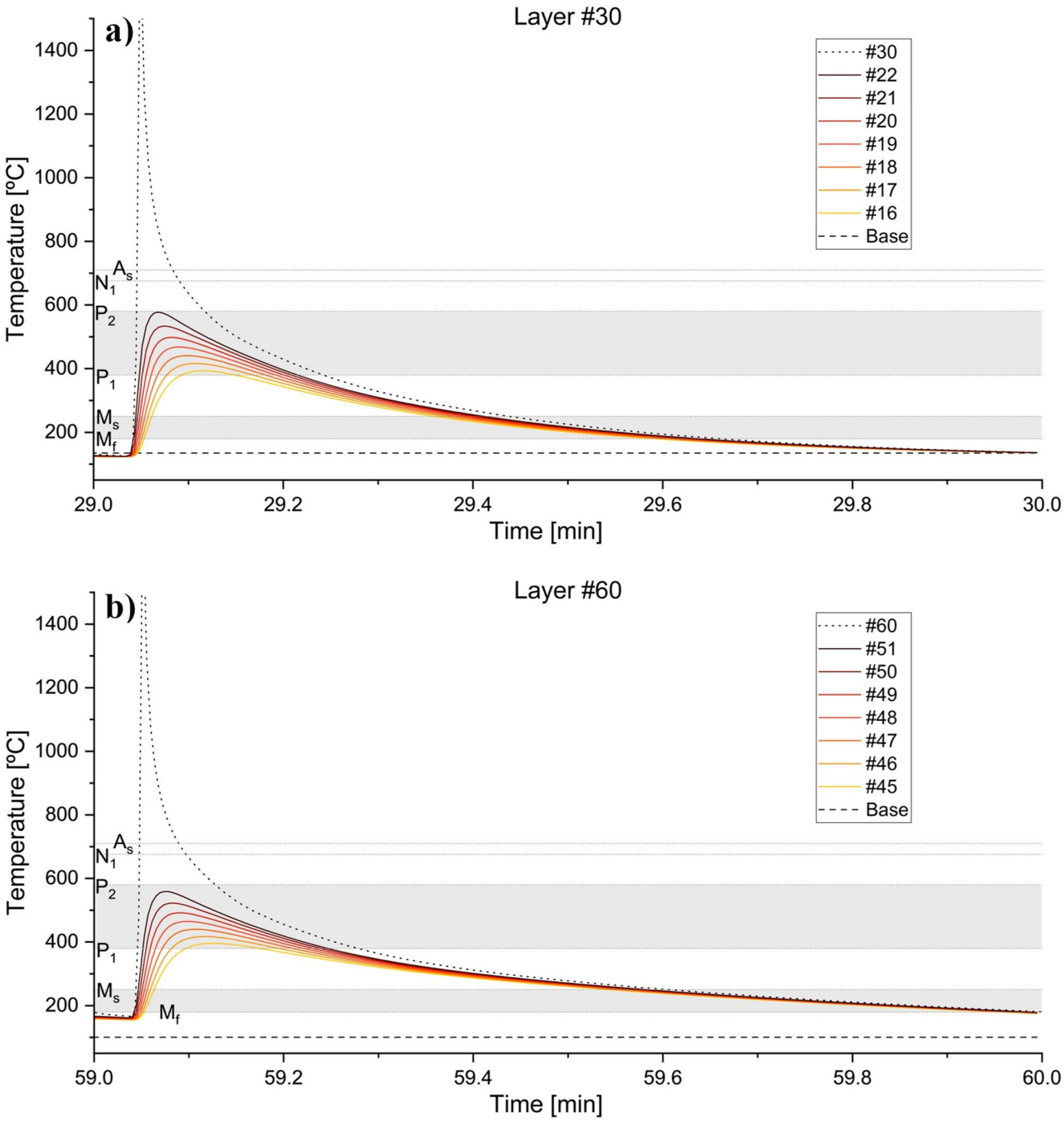


Fig. 12 Layer temperature profiles that meet the aging criterion during the final deposition cycle of walls built with a time between layers of 60 s. The layers that reach a maximum temperature between P_1 and P_2 are selected and drawn in **a** and **b** for the deposition of the 30th and 60th layers, respectively

P_1 and P_2 and for the time spent therein, in accordance with the 390–600 °C aging range obtained from literature for 18Ni maraging alloys [26, 27]. This assumption is justified by the fact that temperatures above 580 °C favor the precipitation of Fe_2Mo at the cost of the redissolution of the main hardening $Ni_3(Ti, Mo)$ precipitates and γ formation, while the reaction kinetics is expected to be slow below 380 °C. An Arrhenius-type or time–temperature equivalence approach was not attempted due to lack of necessary data, so instead the present work relied on a simplified residence time approach.

Table 3 Increase in hardness of walls, ΔH, after different heat treatments

Designation	Average hardness [HV0.1]	ΔH increase [HV0.1]	Aging time [min.]
Single track	343 ± 5		
30L (Hardened region)	395 ± 11	52	0.838
60L (Hardened region)	386 ± 20	43	1
SA+A3h (30L)	559 ± 35	216	180
A3h (30L)	566 ± 26	222	180
A4.5 (30L)	611 ± 31	268	270
SA+A6h (30L)	615 ± 31	272	360
A6h (30L)	624 ± 38	281	360

If the last layer temperature profiles of Fig. 6 are considered quasi-stationary, the simulation shows that the laser heat input causes the underlying material to fulfil the aging condition across 7 consecutive layers, whose temperature profiles are plotted in Fig. 12 and identified by its sequential layer number according to the node height. The set of 7 layers undergoes aging for a cumulative time of approximately 50 s under the 30th layer and 60 s under the 60th layer, which should result in age-hardening of the 18Ni300 maraging alloy, even for these short aging times [28].

It can be noted that the localized γ formation that occurs from P_2 up to A_s (i.e., close to N_1) should take place across a narrow band of material, due to increased temperature gradients. Such γ is therefore considered not to affect significantly the hardness profiles.

The IHT aging times estimated with the FEA model also agree with the age-hardening law that correlates the increase in hardness, ΔH, with the aging time, t. The expression $\Delta H = (Kt)^n$ is known to describe well the hardening kinetics of maraging steels, with the temperature-dependent constants K and n being determined from experimental fitting [29, 30]. Hence, the validation of the model based on hardening kinetics requires measurements of the wall hardness for the extended aging times compiled in Table 3. FEA results show that deposition induces aging for 50 and 60 s in 30L and 60L walls, respectively, leading to hardness increases in the alloy of 52 and 43 HV0.1 relative to a single track. Prolonged aging times of 3, 4.5 and 6 h cause hardness increases by 220, 268 and 275 HV0.1, respectively, regardless of the material thermal history.

Figure 13 shows the experimental data in Table 3 as a plot of ΔH relative to aging time and the best fit to the hardening kinetics law. The fitting demonstrates a good correlation to the hardening law, with $R^2 = 0.991$ for the numerical constants n = 0.294 and $K = 3.360 \times 10^7\ h^{-1}$.

This work uses DTA and thermal inverse FEA of deposition cycles to estimate the laser-induced aging time of a maraging steel during LPD. The results fit a hardening kinetics law with n and K constants that describe the early aging of maraging steels [29]. Knowledge of the precipitation temperature range and the FEA allow to estimate the IHT aging time of the 18Ni300 alloy, and the results of the present study compare well with previous works that describe very similar kinetics with n = 0.29 and $K = 3.1 \times 10^8\ h^{-1}$ [30], where the exponent n matches closely the value obtained in the current study (n = 0.294). Hence, the hardening attained in the deposited walls can be attributed to aging effects occurring during deposition where the thermal cycles induce IHT of the underlying material. These effects can be optimized by adjusting the laser reheating time and the overall temperature cycle.

The current model aims to establish a straightforward FE modelling framework for establishing a connection between

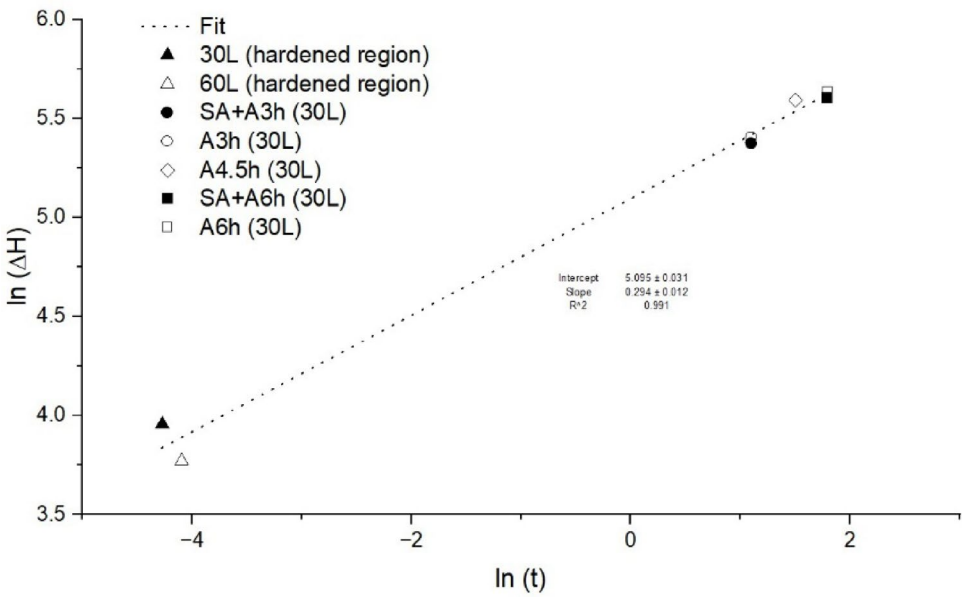


Fig. 13 Increase in wall hardness, ΔH, relative to the aging time of different heat treatments, t

heat transfer conditions and phase transformations occurring in a simple domain, produced by laser powder deposition. This allows for an estimate of intrinsic aging times for the studied conditions. Even in simple geometries, if dimensions differ, or laser parameters are changed, one expects the transient thermal field to change and thus intrinsic aging times, as a result. This is where such a model can be useful in fine tuning parameters like time interval between layers, so as to attain the desired results, namely the promotion of a certain level of aging.

The same rationale would apply to a complex 3D part, wherein printing parameters and trajectories could be optimized by the use of a generalization of such a model, e.g. an inverse modelling algorithm based on the current FE model, with the goal of assuring consistent local aging times, enabling a case-by-case analysis.

4 Conclusions

This work features an inverse FEA model to characterize the IHT of LPD-produced 18Ni300 maraging steel parts, based on deposition temperature measurements and the phase transformation temperatures determined from high-temperature DTA tests. The model reasonably describes the changes in hardness measured along the build direction of thin walls built by LPD. For example, both 30-layer and 60-layer walls show hardness increments of 52 HV0.1 and 43 HV0.1, which the model attributes to IHT times of 50 and 60 s, respectively. These estimates fit the known hardening kinetics of the 18Ni300 alloy, especially its exponent, which shows that this approach: (a) allows for the estimate of precipitation time during deposition and (b) that improved IHT strategies may serve as a future alternative to costly post build-up heat treatments of these alloys.

In short, it was found that:

- The IHT of the thin wall LPD strategy induces aging in 18Ni300 steel;
- DTA results indicate that reaustenitization may occur at temperatures above 710 °C, complete martensitic transformation occurs below 180 °C (M_f), and the precipitation reactions occur between 380 and 580 °C;
- The model estimates that the IHT requires thermal cycles that allow material reheating under 580 °C to avoid precipitate redissolution, and that a set of 7 layers remains in the precipitation temperature range 380–580 °C for 50 or 60 s;
- The estimated IHT aging time fits well the age-hardening kinetics established in the literature, in particular the respective exponent (n = 0.29);
- The determination of a precipitation temperature range, coupled with a simple heat transfer model, appears to be a convenient route for IHT optimization.

Author contributions I.R.: investigation, formal analysis, data curation, writing—original draft. A.M.D.: methodology, formal analysis, writing—review and editing. C.G.F.P.: conceptualization, supervision, writing—review and editing. A.A.: conceptualization, supervision, resources, writing—review and editing.

Funding This work was funded by FCT—Fundação para a Ciência e a Tecnologia (Portugal), under the CeFEMA Project UID/04540/2025 (https://doi.org/10.54499/UID/04540/2025) and the PhD Grant of Ivan Rodrigues SFRH/BD/145658/2019.

Data availability Data sets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

References

1. Hall AM, Slunder CJ (1968) The metallurgy, behavior, and application of the 18-percent nickel maraging steels a survey. Technology Utilization Division, Technology Utilization Division, National Aeronautics and Space Administration
2. Fayazfar H, Salarian M, Rogalsky A, Sarker D, Russo P, Paserin V, Toyserkani E (2018) A critical review of powder-based additive manufacturing of ferrous alloys: process parameters, microstructure and mechanical properties. Mater Des 144:98–128. https://doi.org/10.1016/j.matdes.2018.02.018
3. Jägle EA, Sheng Z, Kürnsteiner P, Ocylok S, Weisheit A, Raabe D (2017) Comparison of maraging steel micro- and nanostructure produced conventionally and by laser additive manufacturing. Materials (Basel). https://doi.org/10.3390/ma10010008
4. ASM Handbook Committee (1990) Properties and selection: irons, steels, and high-performance alloys, vol 1. ASM International. https://doi.org/10.31399/asm.hb.v01.9781627081610
5. Costa L, Vilar R, Reti T, Deus AM (2005) Rapid tooling by laser powder deposition: process simulation using finite element analysis. Acta Mater 53:3987–3999. https://doi.org/10.1016/j.actamat.2005.05.003
6. M. Karg, O. Hentschel, B. Ahuja, D. Junker, U. Hassler, C. Schäperkötter, A. Haimerl, H. Arnet, M. Merklein, M. Schmidt (2016) Comparison of process characteristics and resulting microstructures of maraging steel 1.2709 in Additive Manufacturing via Laser Metal Deposition and Laser Beam Melting in Powder Bed. In: 6th Int. Conf. Addit. Technol., Nürnberg, Germany, https://doi.org/10.13140/RG.2.2.33902.66881
7. De Baere D, Moshiri M, Smolej L, Hattel JH (2022) Numerical investigation into laser-based powder bed fusion of cantilevers produced in 300-grade maraging steel. Addit Manuf 50:102560. https://doi.org/10.1016/j.addma.2021.102560
8. Amirabdollahian S, Deirmina F, Harris L, Siriki R, Pellizzari M, Bosetti P, Molinari A (2021) Towards controlling intrinsic heat treatment of maraging steel during laser directed energy deposition. Scr Mater 201:113973. https://doi.org/10.1016/j.scriptamat.2021.113973

9. C.G. Meacock, Laser Powder Microdeposition of Biomedical Alloys, PhD thesis in Materials Engineering, Instituto Superior Técnico, Universidade de Lisboa, 2010.
10. Tan C, Zhou K, Kuang M, Ma W, Kuang T (2018) Microstructural characterization and properties of selective laser melted maraging steel with different build directions. Sci Technol Adv Mater 19:746–758. https://doi.org/10.1080/14686996.2018.1527645
11. Casati R, Lemke JN, Tuissi A, Vedani M (2016) Aging behaviour and mechanical performance of 18-Ni 300 steel processed by selective laser melting. Metals (Basel) 6:218. https://doi.org/10.3390/met6090218
12. Tewari R, Mazumder S, Batra IS, Dey GK, Banerjee S (2000) Precipitation in 18 wt% Ni maraging steel of grade 350. Acta Mater 48:1187–1200. https://doi.org/10.1016/S1359-6454(99)00370-5
13. Koniorczyk P, Zieliński M, Sienkiewicz J, Zmywaczyk J, Dębski A (2023) Experimental studies of thermophysical properties and microstructure of X37CrMoV5-1 hot-work tool steel and maraging 350 steel. Materials (Basel) 16:1–21. https://doi.org/10.3390/ma16031206
14. Moshka O, Pinkas M, Brosh E, Ezersky V, Meshi L (2015) Addressing the issue of precipitates in maraging steels – unambiguous answer. Mater Sci Eng A 638:232–239. https://doi.org/10.1016/j.msea.2015.04.067
15. Vasudevan VK, Kim SJ, Wayman CM (1990) Precipitation reactions and strengthening behavior in 18 wt pct nickel maraging steels. Metall Trans A 21:2655–2668. https://doi.org/10.1007/BF02646061
16. Jarfors AEW, Matsushita T, Siafakas D, Stolt R (2021) On the nature of the anisotropy of Maraging steel (1.2709) in additive manufacturing through powder bed laser-based fusion processing. Mater Des 204:109608. https://doi.org/10.1016/j.matdes.2021.109608
17. Deus AM, Vilar R (1996) One-dimensional thermal model including the dependence of absorptivity on temperature using Hagen-Rubens equation. Laser Process. Surf. Treat. Film Depos. Springer Netherlands, Dordrecht, pp 195–201. https://doi.org/10.1007/978-94-009-0197-1_10
18. Taylor S, Wright JB, Forrest EC, Jared B, Koepke J, Beaman J (2020) Investigating relationship between surface topography and emissivity of metallic additively manufactured parts. Int Commun Heat Mass Transf 115:104614. https://doi.org/10.1016/j.icheatmasstransfer.2020.104614
19. Wieting TJ, DeRosa JL (1979) Effects of surface condition on the infrared absorptivity of 304 stainless steel. J Appl Phys 50:1071–1078. https://doi.org/10.1063/1.326083
20. Bird RB, Stewart WE, Lightfoot EN (2001) Transport phenomena, 2nd Ed. Wiley, New York
21. S. Bontha, N.W. Klingbeil, Thermal process maps for controlling microstructure in laser-based solid freeform fabrication. In: 2003 Int. Solid Free. Fabr. Symp., Austin, TX, 2003, 219–226. https://doi.org/10.26153/tsw/5557.
22. NiDI, 18 Per Cent Nickel Maraging Steels—Engineering properties, 4419 (1976) 29. https://www.nickelinstitute.org/.
23. L. Costa, R. Colaço, T. Réti, A. Deus, R. Vilar, Tempering effects in steel parts produced by additive fabrication using laser powder deposition. In: Proc. 1st Int. Conf. Adv. Res. Virtual Rapid Prototyp., Leiria, Portugal, 2003.
24. A. Birnbaum, P. Aggarangsi, J. Beuth, Process Scaling and Transient Melt Pool Size Control in Laser-Based Additive Manufacturing Processes. In: Proc. 14th Solid Free. Fabr. Symp. (2003) 328–339.
25. S. Zekovic, R. Dwivedi, R. Kovacevic, Thermo-structural finite element analysis of direct laser metal deposited thin-walled structures, In: 2005 Int. Solid Free. Fabr. Symp., The University of Texas, Austin, Texas, 2005, 338–355. https://doi.org/10.26153/tsw/7093.
26. Yin S, Chen C, Yan X, Feng X, Jenkins R, O'Reilly P, Liu M, Li H, Lupoi R (2018) The influence of aging temperature and aging time on the mechanical and tribological properties of selective laser melted maraging 18Ni-300 steel. Addit Manuf 22:592–600. https://doi.org/10.1016/j.addma.2018.06.005
27. da Fonseca DPM, Feitosa ALM, de Carvalho LG, Plaut RL, Padilha AF (2021) A short review on ultra-high-strength maraging steels and future perspectives. Mater Res. https://doi.org/10.1590/1980-5373-mr-2020-0470
28. ASM Handbook Committee, Heat Treating of Irons and Steels, Vol. 4D, J.L. Dossett and G.E. Totten (Eds.), ASM International, 2014. https://doi.org/10.31399/asm.hb.v04d.9781627081689.
29. L.G. de Carvalho, R.L. Plaut, M.A. de Martorano, A.F. Padilha, Study of Age Hardening Behavior in a 350 Grade Maraging Steel. In: J.S. Carpenter, C. Bai, J.P. Escobedo, J.-Y. Hwang, S. Ikhmayies, B. Li, J. Li, S.N. Monteiro, Z. Peng, M. Zhang (Eds.), Charact. Miner. Met. Mater. 2015, Springer International Publishing, Cham, 2015. https://doi.org/10.1007/978-3-319-48191-3_1.
30. Sha W (2000) Quantification of age hardening in maraging steels and an Ni-base superalloy. Scripta Mater 42:549–553. https://doi.org/10.1016/S1359-6462(99)00394-2

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.